

BAIRNSDALE, Australia

WMO: 949120

Lat: 37.8817S

Lon: 147.5669E

Elev: 163

StdP: 14.61

Time Zone: 10.00 (AUS)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.2	36.3	31.1	25.5	43.7	32.7	27.3	44.2	27.5	55.7	24.9	56.0	5.5	310	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.8	91.8	67.0	86.4	66.5	81.8	65.7	70.4	83.2	68.9	80.1	67.4	77.4	14.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	101.3	72.0	65.8	95.7	71.4	64.1	90.2	70.6	34.4	84.1	33.1	79.9	32.0	77.3	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	22.6	20.0	DB	29.7	103.9	2.2	4.8	28.1	107.3	26.8	110.2	25.5	112.9	23.9	116.4	(4)
(5)			WB	29.2	73.8	2.3	1.9	27.6	75.2	26.2	76.3	25.0	77.4	23.3	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.5	66.7	66.1	63.6	58.6	53.8	49.7	49.1	50.1	53.1	56.2	60.2	63.2	(6)	
	DBStd	8.00	6.56	5.54	5.57	4.87	4.14	3.70	3.59	3.86	4.71	5.43	6.02	6.07	(7)	
	HDD50	181	0	0	0	1	8	50	60	44	15	3	0	0	(8)	
	HDD65	3075	52	45	95	201	347	460	493	461	360	280	172	109	(9)	
	CDD50	2916	518	452	422	259	127	40	32	48	107	196	307	408	(10)	
	CDD65	334	105	77	52	9	1	0	0	0	1	8	29	52	(11)	
	CDH74	3582	1093	649	497	116	7	0	0	1	36	171	389	623	(12)	
	CDH80	1412	506	240	182	27	0	0	0	0	6	48	143	260	(13)	
(14) Wind	WSAvg	9.1	9.4	9.2	9.0	8.0	8.6	8.2	8.9	9.4	9.8	9.7	9.7	9.8	(14)	
(15) Precipitation	PrecAvg	26.6	2.2	2.2	1.8	2.4	1.6	2.8	1.8	1.7	2.3	2.5	2.9	2.5	(15)	
	PrecMax	34.2	5.0	6.1	4.6	7.4	4.2	11.6	4.3	2.8	6.5	4.8	6.3	5.4	(16)	
	PrecMin	17.6	0.6	0.2	0.5	0.4	0.6	0.4	0.5	0.2	1.0	0.6	0.9	0.9	(17)	
	PrecStd	4.3	1.1	1.4	0.9	1.9	0.8	2.8	1.1	0.7	1.2	1.1	1.3	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.7	95.4	92.7	83.8	73.9	67.0	66.3	69.9	78.8	86.1	91.6	96.3	(19)	
		MCWB	69.6	68.7	67.2	64.0	59.2	55.7	53.2	54.9	59.3	63.4	65.0	67.6	(20)	
	2%	DB	92.0	86.9	85.1	76.9	69.2	62.8	62.6	65.4	72.3	78.6	83.7	87.0	(21)	
		MCWB	68.0	67.4	65.5	62.0	57.0	52.9	51.4	52.9	57.3	60.5	64.8	66.5	(22)	
	5%	DB	85.1	81.7	79.1	72.1	65.5	60.0	59.8	62.2	67.7	73.4	77.9	80.8	(23)	
		MCWB	67.4	67.4	64.5	60.3	55.8	51.7	50.2	51.4	55.6	59.0	63.1	65.4	(24)	
	10%	DB	79.1	77.0	74.5	68.2	62.5	57.5	57.2	59.4	63.6	68.0	72.4	75.2	(25)	
		MCWB	66.2	66.2	63.3	59.0	54.4	50.7	49.2	50.3	53.6	57.3	61.5	63.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.4	71.9	70.2	65.3	61.0	57.2	54.9	56.4	61.3	65.6	68.9	70.8	(27)	
		MCDB	90.1	85.4	79.9	77.7	69.3	62.8	61.1	66.8	73.5	77.5	82.2	85.5	(28)	
	2%	WB	70.5	70.2	68.2	63.4	58.5	55.0	53.0	54.2	58.4	62.3	66.3	68.2	(29)	
		MCDB	84.5	81.9	76.9	72.2	66.3	60.2	58.8	62.8	69.4	73.4	78.1	80.9	(30)	
	5%	WB	68.8	68.5	66.4	61.8	56.7	53.2	51.4	52.3	56.2	60.2	64.3	66.3	(31)	
		MCDB	81.0	77.3	75.7	69.2	63.4	58.1	58.1	59.6	65.6	70.0	74.6	77.9	(32)	
	10%	WB	67.1	67.0	64.7	60.2	55.2	51.7	50.1	51.0	54.4	58.0	62.4	64.5	(33)	
		MCDB	77.6	74.5	73.5	66.8	61.7	56.4	56.1	57.9	62.2	67.3	71.2	74.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.8	19.3	20.1	19.2	17.4	16.3	16.8	17.9	19.9	20.7	19.7	20.6	(35)	
		MCDBR	32.2	28.6	29.7	26.6	22.0	19.7	20.5	22.7	27.5	30.4	31.0	31.9	(36)	
		MCWBR	11.7	11.8	12.7	12.9	12.3	11.9	11.9	12.8	14.4	14.5	13.8	13.3	(37)	
	5% WB	MCDBR	26.3	21.9	23.1	21.8	19.5	17.8	17.7	20.5	24.6	26.6	25.9	26.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.397	0.368	0.340	0.330	0.307	0.295	0.294	0.304	0.326	0.340	0.350	0.383	(40)	
		taud	2.378	2.491	2.560	2.562	2.602	2.627	2.619	2.584	2.522	2.506	2.510	2.411	(41)	
		Ebn at Noon	297	298	294	278	266	261	269	283	294	304	310	303	(42)	
	Edh at Noon	Edh at Noon	40	35	30	26	22	20	21	25	30	34	35	40	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2138	1842	1530	1102	777	614	707	977	1350	1732	1969	2190	(44)	
	RadStd	RadStd	141	125	108	76	38	41	37	46	99	112	137	134	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.83	N/A	N/A	+2.59	N/A	N/A	N/A	-185	+304	+131
Regional (27 neighbors)		+0.77	N/A	N/A	N/A	+0.77	N/A	N/A	-184	+265	+99

Nomenclature: See separate page